

Data Provenance and Measurement Audit of the UCI Bike Sharing Dataset

Capital Bikeshare | 2011-2012 | Official UCI archive

Research question: How does hourly bike rental volume vary with weather conditions, working-day status, and hour of day in the 2011-2012 Capital Bikeshare data?

Executive summary

The raw hourly file contains 17,379 rows and 17 columns across 731 dates. The UCI metadata reports no missing values, and the independent programmatic audit found 0 missing cells across 17 columns. Exact duplicates: 0; duplicate date-hour keys: 0; failed range checks: 0; internal consistency failures: 0. The count identity holds for all 17,379 rows.

Observed mean rentals/hour are 204.87 for clear/few-cloud weather ($n=11,413$), 175.17 for mist/cloudy weather ($n=4,544$), and 111.58 for light-rain/snow weather ($n=1,419$). Weather code 4 has only 3 rows on 3 date(s); its descriptive mean is 74.33 and its interval is withheld. Marginal contrasts may reflect calendar and hour composition.

Working-day hourly means peak at 17:00 (525.29 rentals/hour); non-working-day means peak at 13:00 (372.73). The marginal working-minus-non-working mean difference is 11.80 rentals/hour (95% seven-day block interval 3.38 to 19.91). This marginal average hides distinct hourly profiles.

Verdict: suitable for bounded descriptive associations among observed system-hours. Coverage gaps, source-definition conflicts, sparse weather cells and measurement constraints remain material. Causal inference and present-day operational optimization are outside scope.

Research objects

Object	Definition
Stakeholder	Capital Bikeshare operations and planning team; historical operational reference.
Population	Capital Bikeshare system-hours during the 2011-2012 observation period that the dataset is intended to represent.
Sample / unit	17,379 observed rows; one Capital Bikeshare system-hour per row.
Target	cnt: total completed rentals; casual + registered.
Estimand	Conditional mean $E[\text{cnt} \mid \text{weathersit}, \text{workingday}, \text{hr}]$, for observed combinations.

Dataset and provenance

Creator: Hadi Fanaee-T. Repository: UCI Machine Learning Repository. The archived README traces the source to Capital Bikeshare system logs in Washington, D.C., aggregation by the creator into hourly/daily counts, Freemeteo weather information and calendar annotations. Original feed extraction and exact weather alignment were not reconstructed by this audit.

Dataset DOI: 10.24432/C5W894. License: CC BY 4.0, verified via the official-page license link. The archived README additionally asks for citation of Fanaee-T and Gama (2013), Event labeling combining ensemble detectors and background knowledge, DOI 10.1007/s13748-013-0040-3.

Observed period: 2011-01-01 to 2012-12-31. Download: 2026-09-30T02:16:41.598381+00:00 (Singapore date 2026-09-30). No official semantic version is specified; exact bytes define the analyzed version.

Pinned file hashes

hour.csv | 1,156,736 bytes

e03de4ee4ef4dc376ac6e04bf829673c6269e8eba5c60fa121640fa2f829504f
day.csv | 57,569 bytes

a6bcf826782d3c0fbfdcbbead17cd0884185a0daffe8ff10cd48a874ee7ba18be
Readme.txt | 5,607 bytes

b92c8628622948bf0828c43a1b315d1517c3d7fd63654730d0dea379b8e88175
Bike-Sharing-Dataset.zip | 279,992 bytes

b70182d0d0508e9abbb79306ce5c0cec34869000f8220175ac83d11dbe845401

Source disagreements

The webpage reports 17,389 instances; the pinned CSV and archive README show 17,379. Season labels also disagree between the page and archive. Temperature/feeling-temperature definitions differ between min-max normalization on the current page and divisor normalization in the archive. The audit uses numeric season codes and normalized values; no Celsius conversion is asserted.

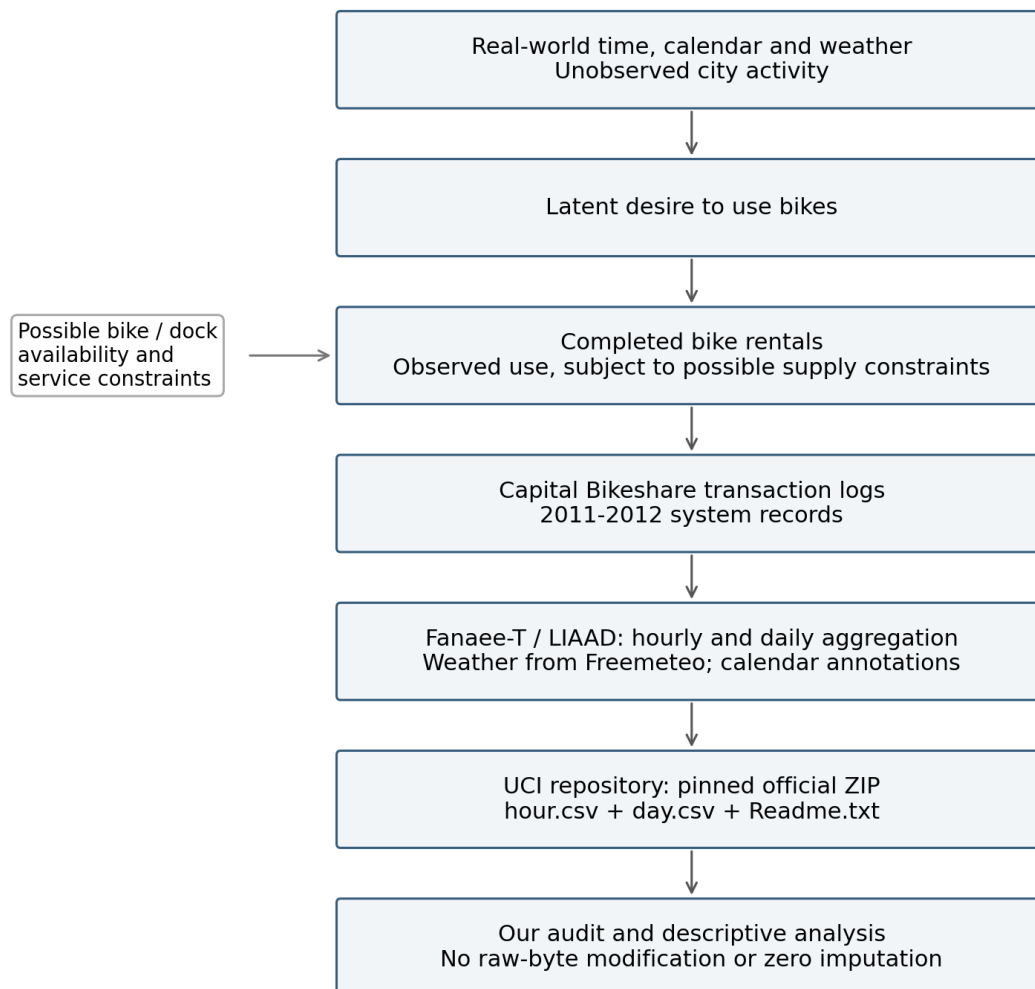
Weekday is described as day of the week, without numeric mapping in either source. All seven cyclic mappings were tested: Sunday=0 fits every row empirically. This is an empirical diagnostic, not an independently documented code definition.

Official page: <https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

Official ZIP: <https://archive.ics.uci.edu/ml/machine-learning-databases/00275/Bike-Sharing-Dataset.zip>

Data-generating process

Data-generating process and provenance lineage



Supply mechanisms are conceptual. Collection, aggregation and weather source follow Readme.txt. Exact original filtering, weather alignment, timezone and DST handling are not documented.

The aggregation and weather source are documented in the raw README. The relationships involving latent demand and supply are a conceptual measurement framework. Exact collection exclusions, original transaction-level processing, clock timezone and daylight-saving treatment are not documented sufficiently to reconstruct them. This audit preserves that uncertainty.

Audit results and data suitability

Check	Executed finding	Meaning
Schema	17,379 x 17; no unexpected/missing columns	Raw schema and dictionary align.
Missing cells	0	Across all raw columns; no blank/text sentinel detected.
Exact / semantic duplicates	0 / 0	Candidate natural key: date + hour.
Range	0 failed checks	Discrete, normalized and count/date domains tested.
Count identity	17,379 satisfy; 0 fail	cnt = casual + registered.
Other consistency	0 failures	Year, month, empirical weekday, workingday, index and day/hour totals.
Coverage	165 absent of 17,544 nominal labels	Observation coverage differs from cell missingness.
Leakage	DETECTED	Outcome components must be excluded from predictors.

A nominal 24-hour-per-date grid contains 17,544 labels; 165 labels have no row. This is record-level incompleteness despite zero missing cells. Clock timezone, daylight-saving handling and omission reasons are undocumented. Absent hours are not assigned zero rentals; estimands are evaluated among observed rows.

Anomaly interpretation

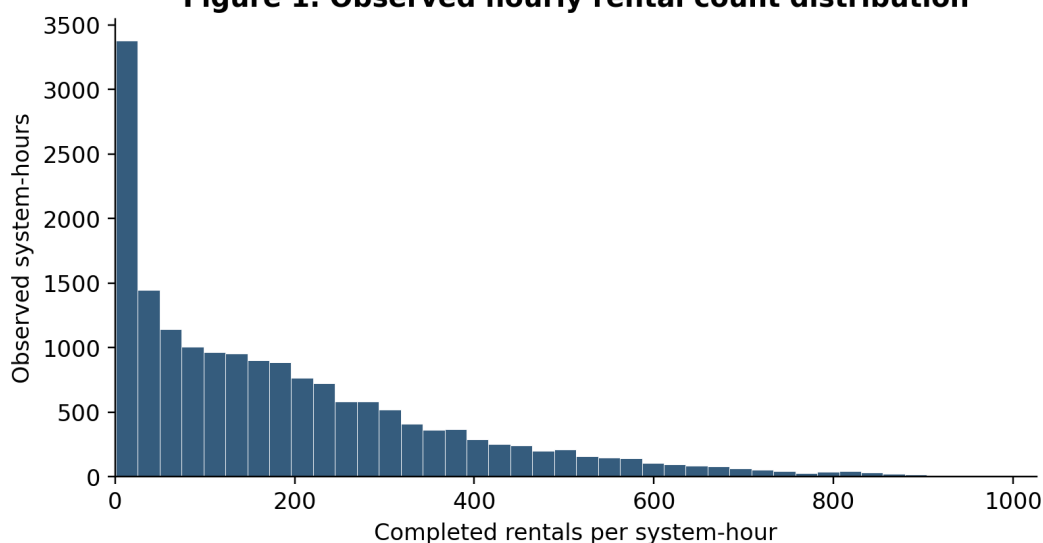
The observed count range is retained in full. Zero-count rows: 0; zero-humidity rows: 22; zero-wind rows: 2180. Zero humidity is a possible measurement concern, not an established missing code. IQR flags and the top 20 counts are saved with calendar/weather context. Statistically extreme rental counts are not automatically treated as data errors because they may represent genuine peak-demand periods.

Variable	IQR flags	Zero rows
cnt	505	0
casual	1192	1581
registered	680	24
temp	0	0
atemp	0	2
hum	22	22
windspeed	342	2180

The largest 20 count records and their hour/workingday/season/weather contexts are preserved in CSV. No unsupported historical-event attribution is made. Zero humidity sensitivity and one-day bootstrap sensitivity are available alongside the main results.

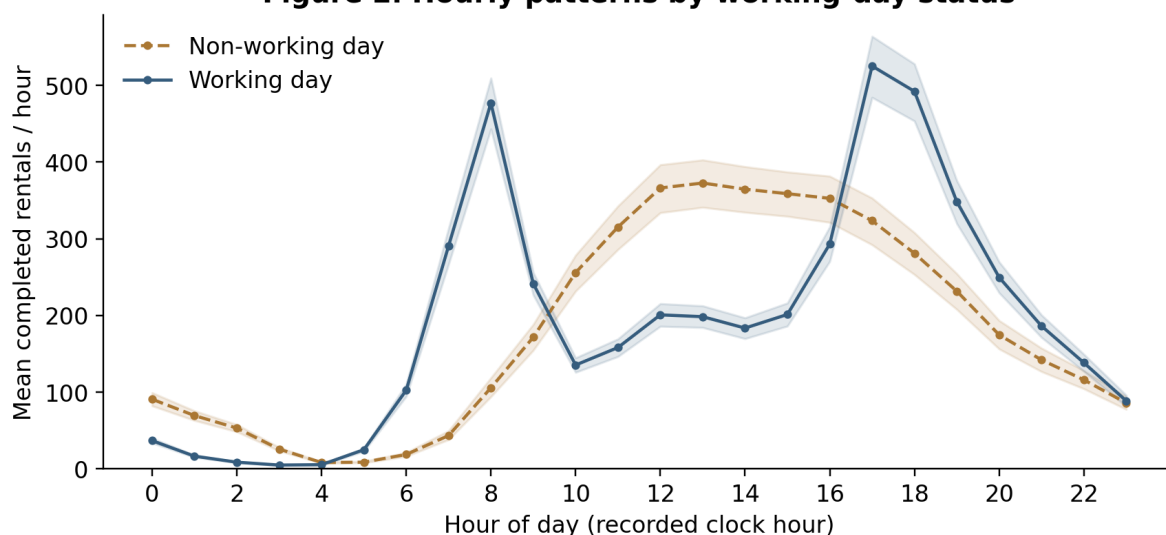
Exploratory analysis: distribution and hour

Figure 1. Observed hourly rental count distribution



Counts are right-skewed; high-use hours are retained rather than removed solely by an IQR rule. The unit is completed rentals per recorded system-hour.

Figure 2. Hourly patterns by working-day status



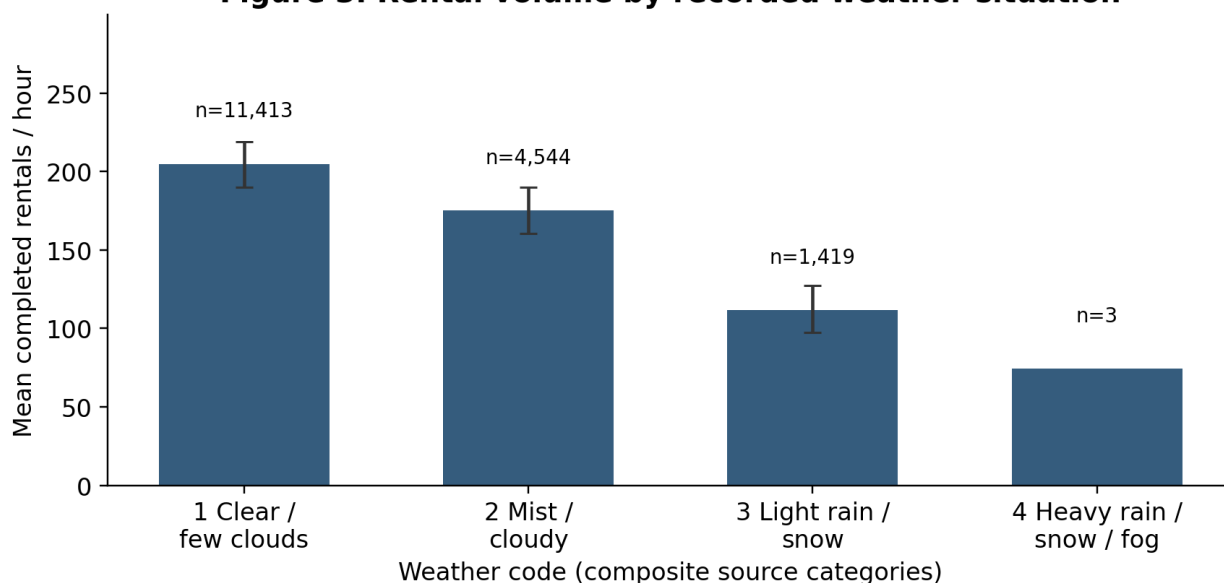
Bands: pointwise 95% seven-day block bootstrap intervals.

Working-day hourly means peak at 17:00 (525.29 rentals/hour); non-working-day means peak at 13:00 (372.73). The marginal working-minus-non-working mean difference is 11.80 rentals/hour (95% seven-day block interval 3.38 to 19.91). This marginal average hides distinct hourly profiles.

Bands are pointwise 95% seven-day moving-block bootstrap intervals. The shape difference contains more information than the marginal working-day mean difference.

Exploratory analysis: weather

Figure 3. Rental volume by recorded weather situation



95% seven-day block intervals; code 4 interval withheld for sparse support.

Weather	n hours / days	Mean	Median / SD	95% interval
Code 1	11,413 / 700	204.87	159.00 / 189.49	189.81 to 219.07
Code 2	4,544 / 567	175.17	133.00 / 165.43	160.73 to 190.01
Code 3	1,419 / 300	111.58	63.00 / 133.78	97.61 to 127.19
Code 4	3 / 3	74.33	36.00 / 77.93	Withheld

Observed mean rentals/hour are 204.87 for clear/few-cloud weather (n=11,413), 175.17 for mist/cloudy weather (n=4,544), and 111.58 for light-rain/snow weather (n=1,419). Weather code 4 has only 3 rows on 3 date(s); its descriptive mean is 74.33 and its interval is withheld. Marginal contrasts may reflect calendar and hour composition.

Inference boundary

Use 2,000 circular moving-block resamples of seven consecutive calendar days, fixed seed 42 (seed offset by block length in code), preserving observed-hour weighting. Intervals are percentile 95%, pointwise rather than simultaneous. They describe temporal resampling stability under approximate assumptions; exact finite-period means are descriptive. Seven-day blocks do not preserve all seasonal dependence or system growth. No probability-sampling or causal interpretation is warranted.

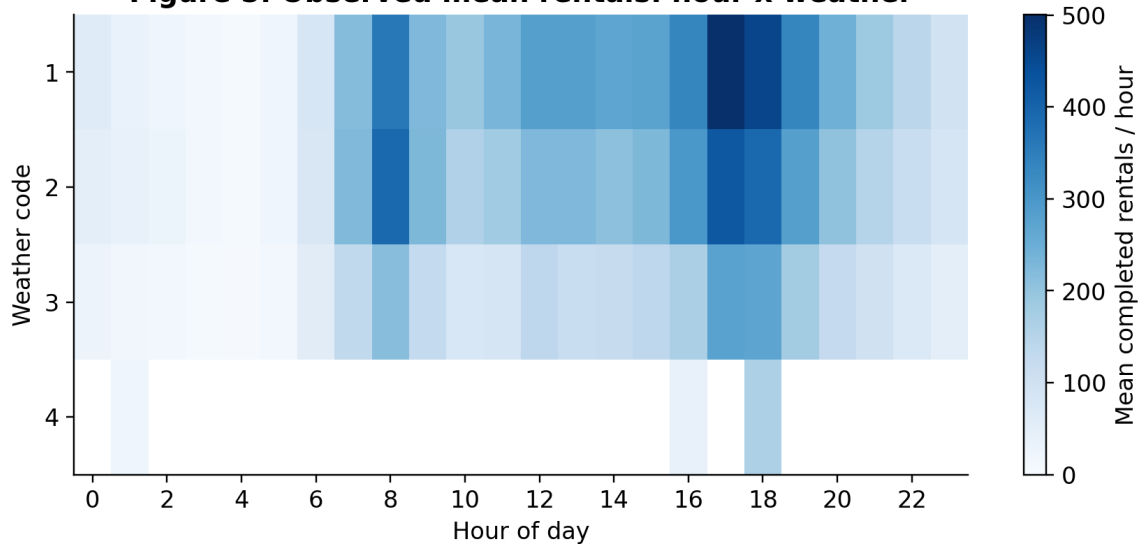
Intervals are withheld below 10 observed days or below 95% finite resamples. One-day cluster sensitivity is saved. Weather categories combine multiple conditions and do not identify a single physical treatment.

Working-day and interaction patterns

Figure 4. Rental distribution by working-day status



Figure 5. Observed mean rentals: hour x weather



Blank cells are unobserved combinations; weather code 4 has sparse support.

The interaction plot summarizes observed weather-hour means, with unobserved combinations left blank. The complete weather x workingday x hour table in `results/conditional_statistics.csv` directly estimates the stated estimand. Each cell records n, observed dates, mean, median, SD and interval status. Sparse cells restrict the supported question; marginal weather patterns can reflect different calendar compositions.

Leakage and measurement validity

Deterministic target leakage

For prediction of cnt, casual and registered constitute deterministic target leakage: cnt = casual + registered on all 17,379 observed rows. Both components must be excluded from predictors. No model is trained. A future-facing model should use chronological evaluation and preprocessing fitted only on the training interval. Same-hour realized weather is potentially post-origin information for advance forecasts; feature timestamps are absent, so forecast availability cannot be certified.

Observed rentals versus latent demand

Conceptual counterexample (not an event observed in these files): 200 people want a bike, only 100 bikes are available, and 100 rentals are completed. Observed cnt = 100 while latent demand is approximately 200. Bike availability, dock capacity, service outages and supply constraints are possible mechanisms. The CSV has no direct measures of unmet demand or these constraints; cnt measures completed rentals.

Bias and limitations

The temporal boundary is 2011-2012 and the system boundary is Capital Bikeshare in the Washington, D.C. area. Season, calendar, year and hour may confound marginal weather associations; system growth and serial dependence also limit statistical interpretation. Sparse weather-hour-workingday cells do not identify all conditional means reliably. Missing system-hours and measurement constraints limit representativeness. Historical patterns do not establish optimal rebalancing or current operating decisions.

Source and coverage qualifications

Data integrity does not establish complete observation or valid causal identification. Humidity boundary values, absent hour records and conflicting metadata are retained as explicit qualifications. The auxiliary daily file validates rental totals, but does not prove every original system-hour was recorded. Original transaction-to-aggregate filtering and weather matching remain unknown.

Prohibited claims

1. Causal weather effects

We must not claim that weather conditions causally determine bike rental demand based on this observational dataset. Weather is not randomized; season, hour, working-day status and calendar patterns can confound the association.

2. Universal or present-day generalization

We must not generalize the observed relationships to all cities, all bike-sharing systems, or current populations. The sample covers a single historical system and transportability has not been tested.

3. Perfect latent-demand measurement

Observed rental counts should not be interpreted as a perfect measurement of unconstrained latent demand. Completed rentals can be supply-limited and the files lack direct measurements of unmet demand.

Conclusion

The pinned data support a testable descriptive question about conditional mean completed rentals among observed 2011-2012 Capital Bikeshare system-hours. Hourly profiles differ by working-day status and rental volume differs across recorded weather categories. Data-level integrity passes do not remove source-document conflicts, absent-hour coverage, sparse cells, confounding or imperfect measurement. No causal, universal or optimal-dispatch claim follows.

Reproducibility

Executed on Windows-10-10.0.26200-SP0, Python 3.11.5. Exact direct package versions are pinned in requirements.txt and environment.yml. The full pipeline verifies raw hashes, executes the notebook in a fresh kernel, generates this report and independently validates files, statistics and claims. Run from the project root:

```
python -m pip install -r requirements.txt
python scripts/run_all.py
```

Final fresh-kernel rerun and verification are recorded in results/qa_validation.json. The pipeline exits nonzero on failure. All paths are project-relative; no private machine path is required by project source. PDF creation does not require a TeX installation. Repository creation and push are separate from the analysis pipeline.

Verified project repository: <https://github.com/Zhoujie-SONG/DS-A1-bike-sharing-audit>

AI use and source attribution

OpenAI Codex / ChatGPT-assisted workflow created the project, audit/statistical code, notebook, plots, report and documentation, then ran source/result checks. The user supplied the fixed specification. No instructor or second human review is claimed. Accepted instructions and Codex-modified implementation choices are disclosed in docs/ai_use_log.md. Programmatic checks and Codex visual review are separately identified in the QA record.

Key implementation decisions include time-block bootstrap instead of independent hourly resampling, interval withholding for sparse support, no unsupported Celsius conversions, and empirically labeled weekday mapping. The student must inspect the submitted work and follow their course disclosure policy.

Sources actually used

Fanaee-T, H. (2013). Bike Sharing [Dataset]. UCI Machine Learning Repository.
<https://doi.org/10.24432/C5W894>

Official dataset page (archived with SHA-256):
<https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

Unchanged official ZIP:
<https://archive.ics.uci.edu/ml/machine-learning-databases/00275/Bike-Sharing-Dataset.zip>

Readme.txt included in that ZIP, describing the creator, system-log aggregation, Freemeteo weather, fields and original citation request. The original system/weather feeds were not directly accessed for this audit.

CC BY 4.0 license: <https://creativecommons.org/licenses/by/4.0/>

README-requested paper attribution: Fanaee-T, H. and Gama, J. (2013). Event labeling combining ensemble detectors and background knowledge. Progress in Artificial Intelligence.
<https://doi.org/10.1007/s13748-013-0040-3> (bibliographic details reproduced from the official README; paper not independently analyzed).

Companion evidence

Inspect the executed notebook, source_metadata.json, audit_results.csv, audit_summary.json, descriptive_statistics.csv, conditional_statistics.csv, statistical_method.json and raw-data dictionary. These preserve negative results, sparse-cell status and the actual dataset version rather than relying on this narrative alone.

Submission text is provided separately in reports/submission_summary.txt. Its repository URL is populated from a verified creation receipt when available; no repository address is invented.